A Household in Transition: A Study on Dynamics

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***Abstract*—Intra-household bargaining power has been shown to increase with a positive non-labor income shock to women. I propose a new method to study female empowerment as a result of the receipt of a cash transfer. Previous research on dynamics has relied solely on cross-sectional analysis. Using data from a cash transfer experiment in Malawi and Liberia, I analyze the dynamics using impulse response function estimation and dynamic multipliers. This paper is therefore the first to estimate impulse response functions of intra-household bargaining power using high-frequency experimental panel data. In doing so,**

***Index Terms*—Collective household, dynamic causal effect, intra-household bargaining, local projections, household dynamics**

I. Introduction

Conditional cash transfers are thought to be an effective policy in alleviating poverty and can be instrumental in reducing gender inequality. The reasoning is that, by increasing her income, a woman's bargaining power increases within her household and therefore her empowerment. An increase in non-labor income specifically has been shown to affect a woman's bargaining power since it changes her relative income, resulting in a better bargaining position. Through this channel, the World Bank and other organizations target cash transfers to women in developing countries with the goal of reducing gender inequality (Hanmer et al. 2014).

We are therefore interested in the following questions: (i) What is the effect of a positive change in non-labor income on women's bargaining power in the current period? (ii) In the next period? and (iii) Into the indefinite future?

Women's bargaining power will be measured as their *resource share* (Browning, Pierre-Andre Chiappori, and Lewbel 2013), which maintains the collective household assumption of unique utility functions. These resource shares will be identified by comparing Engel curves of goods consumed exclusively by gender (Dunbar, Lewbel, and Pendakur 2013) and (Calvi 2020).

Shifts in bargaining power have been studied as a result of cash transfers and other kinds of income shocks (Bobonis 2011), but the dynamics and general transmission mechanisms resulting from any income shock are understudied. While there has been some cross-sectional work (Aggarwal et al. 2022) on intra-household bargaining over time and analysis completed in an intertemporal setting (Mazzocco 2007; Lise and Yamada 2019), the actual transmission and overall dynamics within the system as a whole are, to the best of my knowledge, yet to be analyzed to their fullest extent.

Therefore, the contribution of this paper is to provide a new analytical framework using local projections to study intra-household bargaining, through which we may gain insights into potential structural effects created by such targeted policies.

I use a randomized control cash transfer program conducted in Malawi and Liberia to estimate the dynamic effects on intra-household bargaining power from receiving a cash transfer.

In doing so, this analysis will help clarify how cash transfers influence individual-level outcomes such as labor supply, health, and education (Doss 2013), not only in the short term but in the long run as well. With this framework, it may be possible to more precisely target policies that provide not only temporary improvements in gender equality but lasting structural change.

II. Review of Literature

A. The Collective Household

The collective household model, originally introduced by Browning and Pierre-André Chiappori 1998, assumes that households are Pareto efficient, meaning no resources are wasted.

This model was developed after several empirical tests rejected the unitary household model (Becker 1973), in which households were shown not to be Pareto efficient and provided evidence against the income pooling hypothesis. These studies showed that a mother's unearned income significantly impacts child health more than fathers' (D. Thomas 1990) or results in more spending on women's and children's clothing (Lundberg, Pollak, and Wales 1997), suggesting that distinct individual preferences exist and that bargaining power matters. Thus, the collective household model was adopted to model this intra-household bargaining process.

B. Non-Labor Income and Intra-Household Bargaining

Using the collective household model, it has been shown that changes in relative income can significantly affect labor or educational outcomes. Receiving non-labor income, in particular, was found to be a driver of significant shifts in bargaining power. Changes in pension incomes (Duflo 2000) or conditional cash transfers (Bobonis 2011) can result in such shifts.

The literature finds that cash transfers are significant in increasing a woman's Pareto weight (Adato et al. 2000, De Rock, Potoms, and Tommasi 2022). We can therefore start from the assumption that a positive income shock in the form of a cash transfer increases a woman's bargaining power and therefore seek to qualify its adjustment process.

C. Dynamic Models and Interpretations

Dynamic analysis in household models has been conducted via intertemporal choice models (Mazzocco 2004, Ligon,

J. P. Thomas, and Worrall 2002), and general cross-sectional analysis (Aggarwal et al. 2022) has been used to analyze dynamics within households. The authors in these papers use a Limited Information Commitment (LIC) model for their analysis. The closest research to what I am proposing is by Aggarwal et al. 2022, in which they measure indices and estimate cross sections over time. In fact, the data collected by this group of researchers are the data proposed to be used in this project.

Although significant additions to the literature, the existing methodologies do not address the question of the exact adjustment process. While they provide excellent references, I therefore turn to a common applied macroeconomic analysis tool.

Using time series analysis in an empirical development setting, specifically for panel data, has previously been proposed in an educational context, such as in Kaplan 2002. The author argues for the use of dynamic multipliers in panel data becasue

A popular method of estimating impulse response functions is via local projections (LPs). Although there are bias–variance trade-offs in their usage compared to typical VAR estimation (Montiel Olea et al. 2025), local projections do not require full knowledge of the underlying distribution of the estimator (Jordà 2005) and are therefore better suited to this context.

In addition, VAR estimation requires stationarity, which in this context could be met by assuming that the household is one of commitment. However, more recent literature on household bargaining models indicates that marriage, and therefore bargaining power, is in a constant state of renegotiation, implying a model that varies over time and violates this condition. Local projections do not require stationarity (although it may be helpful), which is another advantage of this methodology.

III. The Data

Researchers partnered with Innovations for Poverty Action, GiveDirectly, and USAID to conduct a randomized evaluation to measure the impact of an unconditional cash transfer and market access program on a set of outcomes. These outcomes included food security, spending, income, resilience to health shocks, intimate partner violence against women, and psychological well-being (Aggarwal et al. 2022).

A. The Setting

Malawi is a small country in southeastern Africa. Culturally, the south is mainly matrilineal and the north is more patrilineal, while the central region includes both matrilineal and patrilineal tribes (Zahra, Kidman, and Kohler 2021).

Liberia is a small country on the west coast of Africa. Marriage rates in rural areas are much higher than in urban areas.

Both countries are considered low income with high rates of poverty. In 2016 (the most recent data available), over half of the population in each country lived below the poverty line (50.9 percent for Liberia[1] and 50.7 percent for Malawi[2]. Therefore, a cash transfer program might allow for more flexibility in spending and cause budget constraints to be non-binding.

Culturally, both Malawi and Liberia have very high divorce rates but appear to be decreasing and stabilizing over time (Clark and Brauner-Otto 2015). With divorce being so common, we can assume that individuals residing together remain together because their current arrangement is better than their outside option. This therefore supports the plausibility of participation constraints being satisfied on average. and may employ a cooperative household model as we do here.

B. The Experiment

This experiment was originally conducted to consider whether unconditional cash transfers can have persistent effects on other dimensions of household welfare (Aggarwal et al. 2022). Therefore, the data include information on household consumption, transfers, market values of various goods, as well as demographic information such as years of education, gender, and psychological well-being[3].

In total, 600 villages were included in this experiment, split equally between Liberia and Malawi. Within each country, half of the villages were targeted to receive cash transfers, while the remaining 300 served as controls. Payments were in 250 dollars, 500 dollars, or 750 dollars per household. These sums are paid out in monthly 250 dollar increments, so that villages receive 1, 2, or 3 payments.

Within each household, the cash transfer was made to a beneficiary chosen by the household. The majority of beneficiaries were women, but the choice was largely endogenous.

Of the entire sample, 20 percent were randomized into phone surveys to collect information on expenditures, labor supply, and related outcomes. The frequency of data collection was such that each household was called every other month, with households within a village alternating months. Therefore, each village has a data point for every month. Monthly prices from 80 markets in Liberia and 95 in Malawi were also collected over a two-year period, starting before the transfers began.

This dataset therefore provides a rich source of high-frequency data with relevant information on demographics and consumption patterns.

IV. Methodology

In this project, I seek to estimate the dynamic causal effects—or impulse response functions—of cash transfers on female empowerment. To do so, I obtain a proxy for female empowerment as her Pareto weight using the framework of the collective household model. The weight will be estimated using Engel curves. I then construct a time series of these Pareto weights and analyze their response to an income shock.

[1] [https://pip.worldbank.org/country-profiles/LBR

[2] https://pip.worldbank.org/country-profiles/MWI

[3] https://www.socialscienceregistry.org/trials/4869

A. The Collective Household Model

Following the collective household model as in Browning and Pierre-André Chiappori 1998, we define the household's problem as

$$\max_{c^i} \left(1-\theta^f\right) u^m \left(c^m, c^f, c^h\right) + \theta^f u^f \left(c^m, c^f, c^h\right) \tag{1}$$

subject to

$$p \cdot \left(c^m + c^f + c^h\right) \leq y^m + y^f + y^h \tag{2}$$

In Equation (1), $u^j(\cdot)$, $j \in male, female$, represents the utility function of either the male (husband) or female (wife), and c^i, $i \in male, female, household$, represents either gendered consumable goods (such as suits for men or dresses for women, although this need not always be the case) or common goods for the household. Equation (2) represents the budget constraint, specifying income to the man, woman, or the household as a whole.

B. Estimating the Pareto Weight

Ultimately, I am interested in estimating the woman's Pareto weight θ^f, a measure of women's bargaining power. However, most consumption data are not indexed to the individual, so we employ the method used by Calvi 2020.

Within the context of a cooperative household model (as is used here), the following is taken to be true:

Proposition 1: If a complete system of demands for goods is observed, and private good 1 (resp. 2) is exclusively consumed by member A (resp. B), the indirect collective utility functions are identified up to a monotonic transformation. For any choice of cardinalization, the function is exactly identified. (Pierre-André Chiappori and Donni n.d.)

From this idea, we rely on previous results from Browning, Pierre-Andre Chiappori, and Lewbel 2013 that, for a general linear technology, it is possible to identify the decision-making prices (i.e., the Pareto weights) when estimating the demands of singles and couples, defined as resource shares.

Resource shares are then identified by comparing Engel curves of assignable clothing items—goods consumed exclusively by women, men, or children. The data provide weekly consumption data and prices for all 300 villages included in the study, allowing for such a comparison.

C. Incorporating Dynamic Causal Effects

Local projections (LPs) are a time series methodology typically used in empirical macroeconomics, for example in studying bank-loan balances Dassatti et al. 2024. I find this approach applicable in a development context because it allows for analysis beyond static comparisons.

Requirements that the shock be unanticipated can be satisfied because this experiment is a randomized control trial and therefore receipt of the cash transfer is unexpected to the household and exogenous.

Using local projections (LPs) as defined in Jordà 2005, the following regression will be run for each time horizon $h = 0, 1, \ldots$:

$$\theta^f_{t+h} = \mu_h + \lambda^{\mathrm{LP}}_h x_t + \gamma'_h r_t + \sum_{\ell=1}^{p} \delta'_{h,\ell} w_{t-\ell} + \xi_{h,t}. \tag{3}$$

Here, θ^f_{t+h} is the woman's bargaining power h horizons ahead from time t; x_t is a (scalar) impulse variable i.e. the cash transfer, r_t is a vector of time series that is included as contemporaneous controls like current age, length of marriage, socioeconomic status, education level of the woman, education level of the husband, etc; $w_t = (r'_t, x_t, \theta_t, q'_t)'$ collects all variables included as lagged controls, with q_t a potential additional control vector, and $\xi_{h,t}$ is the multi-step forecast error in the regression.

By the Frisch–Waugh–Lovell theorem, the regression coefficient λ^{LP}_h can be interpreted as the coefficient from a projection of the outcome θ^f_{t+h} on a "shock" $\tilde{x}_t$, defined as the residual from a projection of x_t on the control variables in Equation (3) (Montiel Olea et al. 2025).

After estimating θ^f_t for every time period t, the dynamic multiplier λ^{LP}_h provides an estimate of the persistence, or lasting effect, of a cash transfer on a woman's bargaining power within her household.

V. Implications

Analyzing the dynamics of household models via dynamic multipliers addresses a critical policy question: how long does it take for a policy change (such as Progresa or other cash transfer programs) to generate a substantively important effect on female empowerment within the household? With this study, it may be possible to better determine policy effectiveness by assessing the degree of structural change achieved.

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